A few notes...

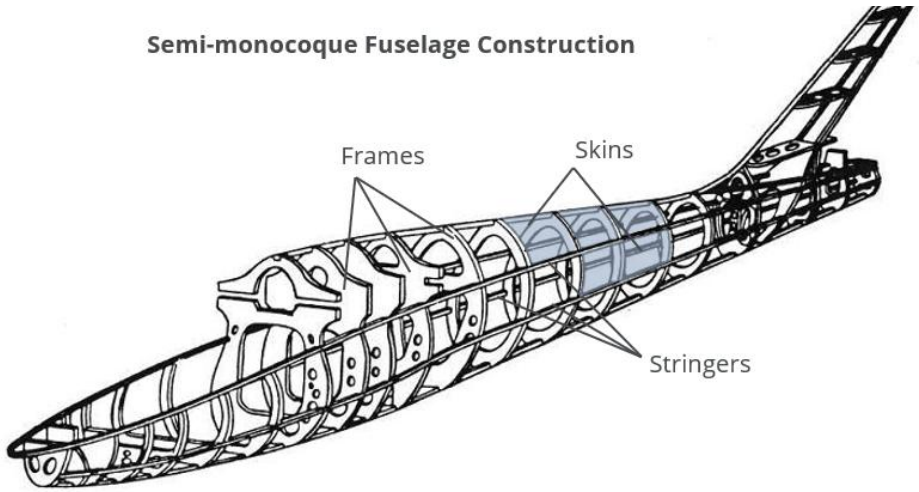
- **HW6 is due 11:59pm TONIGHT**

- **NASTRAN/FEMAP assignment (which we will call HW7) is officially assigned today and due 11:59pm Wed Apr 7**

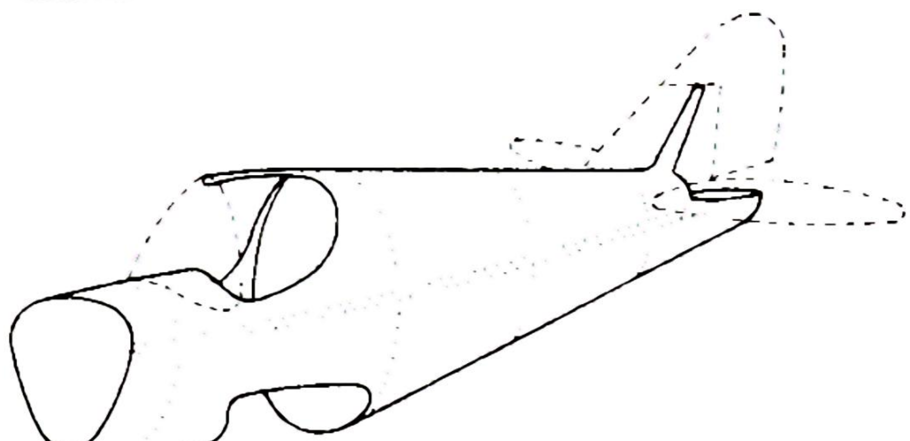
- **Consult Anthony Seholm with any questions on this assignment (seholma@my.erau.edu, office hrs are MWF 1-3, TTh 11-1 in LB243)**

Chapter 5: Wings and Fuselages with Structurally Significant Skin

- **In this Chapter we analyze wings and fuselages with "structurally significant" skin, i.e. skin that is stiff enough to carry a significant portion of the load**

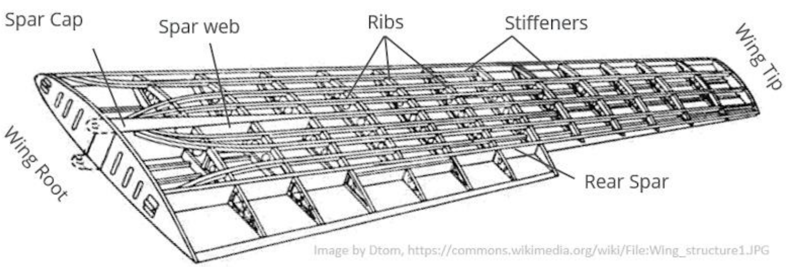


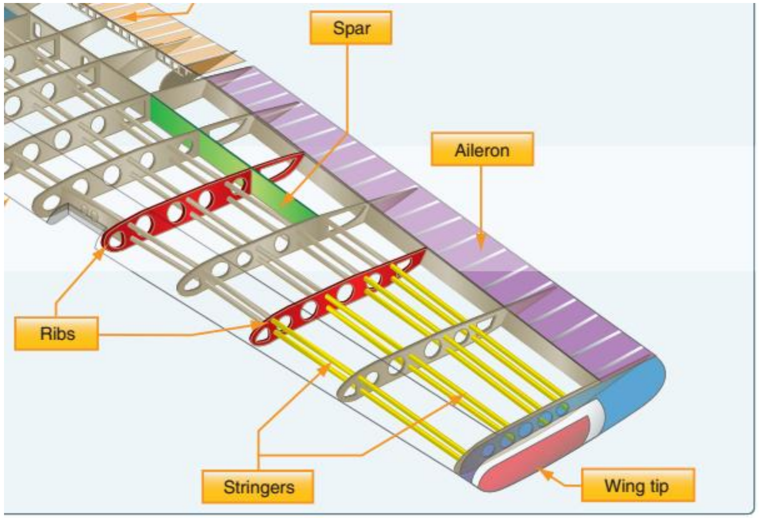
- **A semi-monocoque structure is comprised of structural elements that carry the loads and stress on an aircraft**



- **A fully monocoque structure consists of a design in which only the skin of the aircraft carries the flight loads**

Chapter 5: Wings and Fuselages with Structurally Significant Skin





- **Flanges (Spar Caps)**

These consist of the upper and lower flanges attached to the spar webs. The flanges carry the bending moment generated by the wing in flight. The flanges also support the wing skin against buckling.

- **Spar web**

The spar web consists of the material between the flanges and maintains a fixed spacing between the them. The spar web is responsible for carrying the vertical loads (lift).

- **Wing Ribs**

The ribs help to maintain the aerodynamic profile of the wing. The ribs form part of the boundary onto which the skins are attached, and support the skins and stiffeners against buckling.

- **Stringers (Stiffeners)**

Stiffeners or stringers form a part of the boundary onto which the wing skin is attached and support the skin against buckling under load. The stiffeners also carry axial loads arising from bending moments in the wing.

- **Skin**

The wing skin transmits in-plane shear loads into the surrounding structure and gives the wing its aerodynamic shape.

5.1: Idealization of Beam Cross-Sections

- **Complex semi-monocoque structures such as wings and fuselages are often idealized to simplify their analysis**

- **The usual assumptions are:**
  - 1. The stringers (stiffeners) carry axial stresses (σ), and these axial stresses are constant over the cross-section of the stringer ($\sigma = P/A$)
  - 2. The thin web (essentially the skin) carries shear stresses (τ), and these shear stresses are constant over the thickness of the web
  - 3. The ribs are rigid within their own planes, so that the cross-section is maintained unchanged during loading; however, they are assumed to possess no rigidity normal to their plane, so that they offer no restraint to warping deformations out of their plane

- **Assumptions 1 and 2 present some difficulty since the webs can carry a significant amount of the axial stresses as well as shear stresses. Thus, it is necessary to use an analytical model which takes into account the fact that the web cross-section can carry some normal (i.e. axial) stresses**

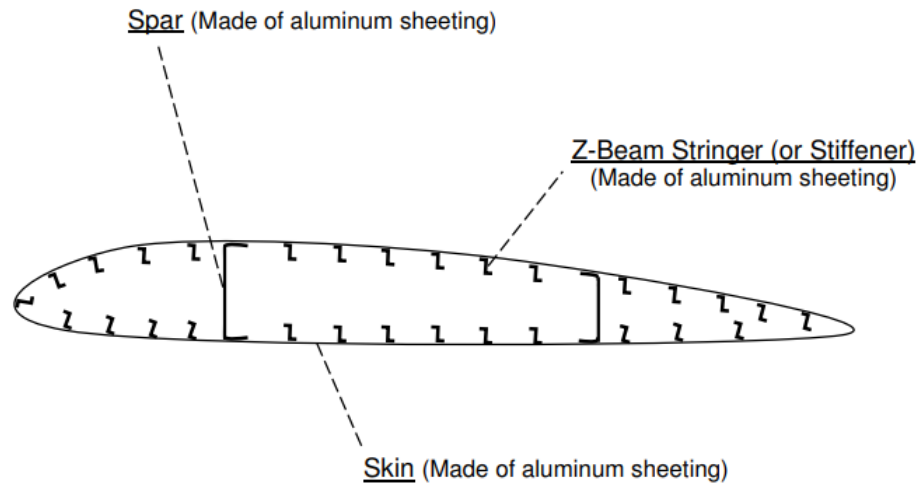
- **This is done via the concept of an "ideal" stringer, i.e. one whose cross-section is larger than the actual stringer cross-section**

5.1: Idealization of Beam Cross-Sections

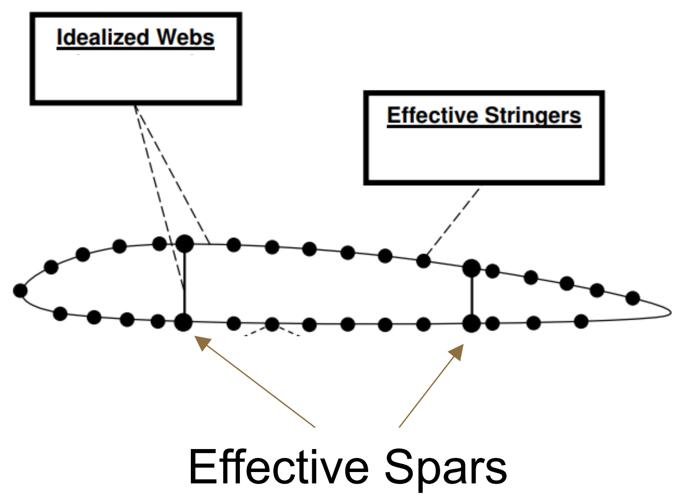
- The larger cross-sectional area is called the effective stringer area
- Its numerical value is determined by "lumping" some of the web cross-sectional area that is adjacent to the stringer with the actual stringer cross-sectional area, i.e. a lumped-mass model
- The effective stringer area is then placed at the centroid of the new area, along the centerline of the web
- The effective stringers are then assumed to satisfy Assumption 1 and the webs are assumed to satisfy Assumption 2
- Therefore, the resulting lumped-mass (idealized) cross-section for the assembly will satisfy the assumptions above

5.1: Idealization of Beam Cross-Sections

- **This figure represents an actual wing construction → the entire cross-section collectively carries both normal and shear stresses**

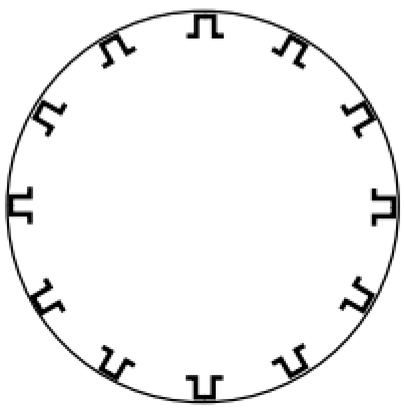


- **This figure shows the same wing cross-section idealized for analysis → the effective stringers carry normal stresses and the idealized webs carry shear stresses**

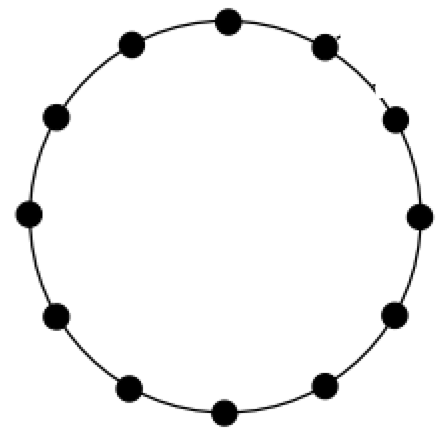


5.1: Idealization of Beam Cross-Sections

- **Here are similar figures for a fuselage with stiffeners:**



Actual Fuselage Construction
[Entire Cross-section carries both σ and τ]

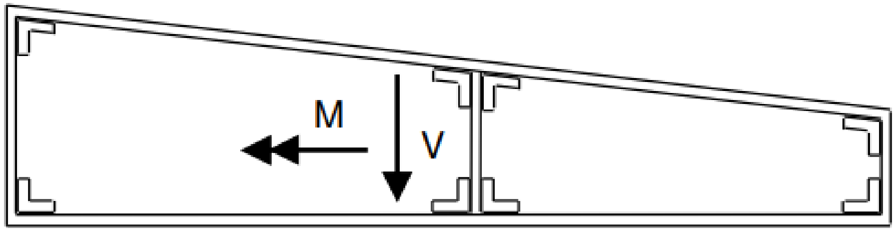


Idealized Fuselage Cross-Section
[Only Effective Stringers Carry σ]
[Idealized Webs Carry Only τ]

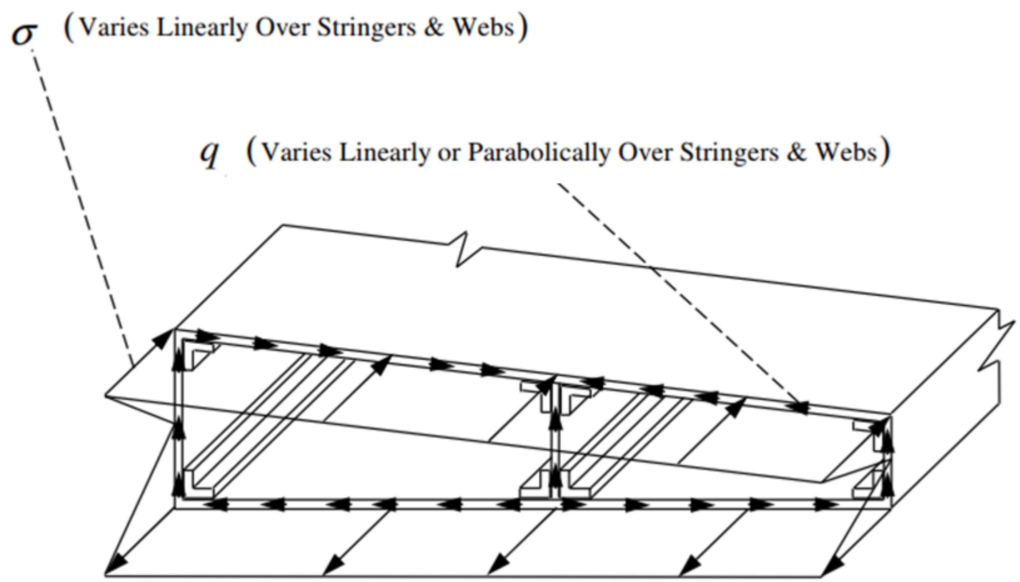
NOTE: The textbook tends to use the term "stringer" and "stiffener" interchangeably, as well as the term "web" and "skin"

5.1: Idealization of Beam Cross-Sections

- **Consider the actual cross-section for a wing portion, having an internal bending moment *M* and internal shear force *V***



- **The bending stress distribution *σ* and shear flow distribution *q*, (which correspond to *M* and *V*, respectively) would look something like this:**



[Both σ and q act over the entire cross-section]

Aerospace Engineering Department
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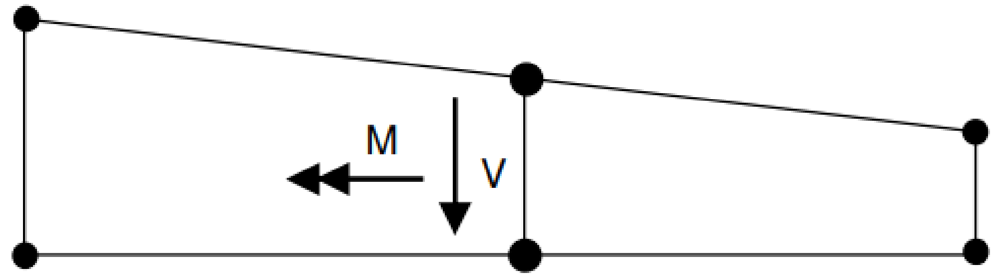
Daytona Beach Campus

5.1: Idealization of Beam Cross-Sections

- **Idealizing the wing section Looks as shown here, with effective stringers and idealized webs**



- **The bending stresses and shear flows, which correspond to *V* and *M* above, would look something like this:**